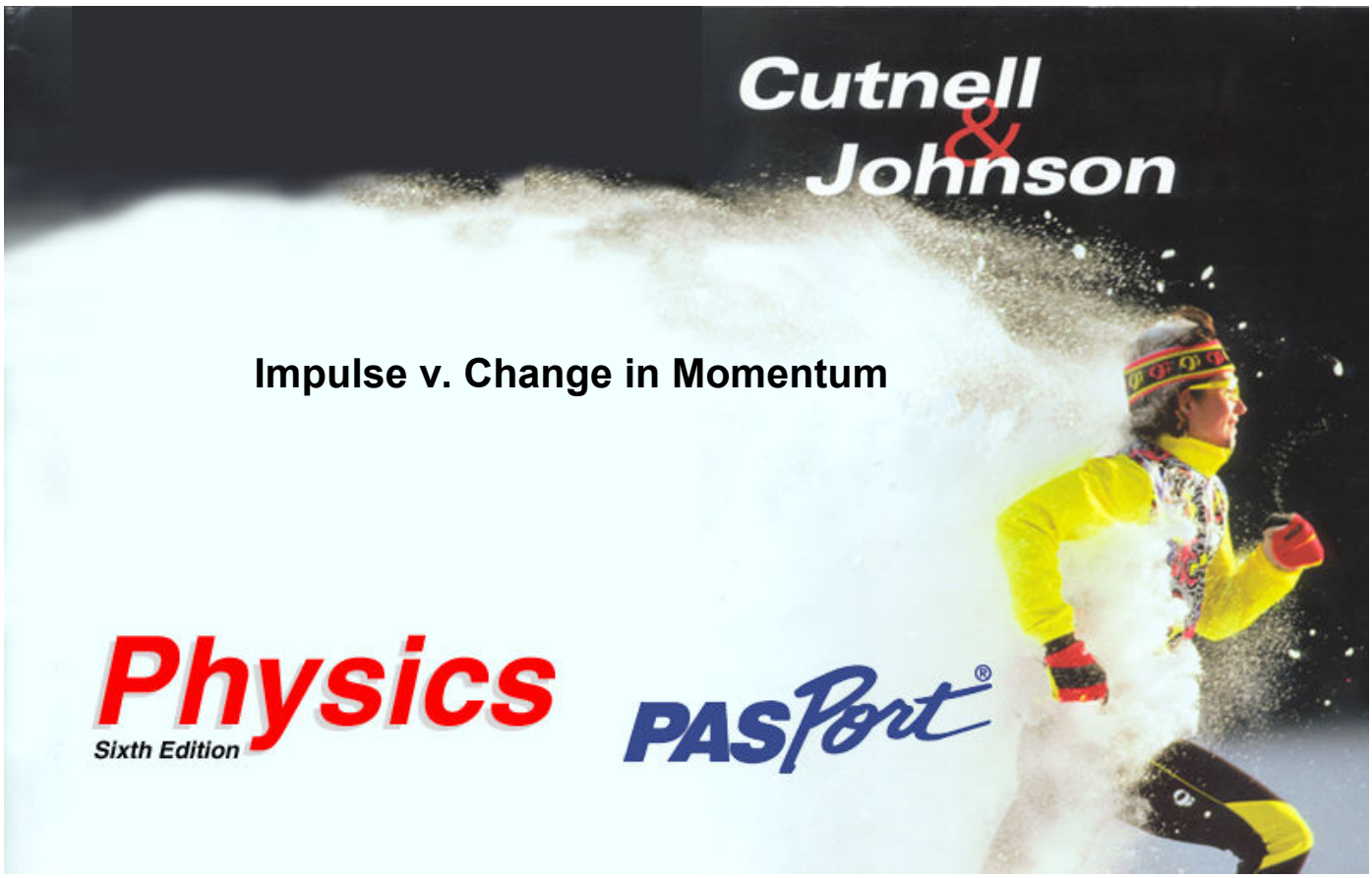

Cutnell & Johnson
Impulse v. Change in Momentum
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

Date:	
Your Name:	
Group:	
Period:	

Introduction

Exploration Goals

§ In this exploration you will determine the relationship between the change in momentum and the impulse (net force multiplied by time).

§ Use the Motion Sensor to measure the motion of a cart as it collides with a magnetic bumper.

§ Use a Force Sensor mounted on the track to measure the force of the collision over the same interval of time. Compare the change in momentum of the cart with the area under the curve of the measured force vs. time graph.

Learning Outcomes

You will be able to:

- Understand the relationship between impulse and change in momentum.
- Use appropriate technology to measure the change in velocity and the force when a cart collides with a bumper.
- Calculate the change in momentum using the mass and the change in velocity.
- Calculate the impulse using the force and the time interval.
- Recognize that the area under a plot of force vs. time for a collision is equal to the impulse.

Hypothesis

§ How is the amount of impulse exerted on an object during a collision related to the change in momentum of the object during the collision?

Background

§ The impulse of a force is the product of the average force and the time interval during which the force acts. (See equation 8.1.) Impulse is a vector quantity and has the same direction as the average force. The SI unit of impulse is the newton·second (N·s).

$$\textbf{Impulse} = \boldsymbol{F}\Delta\boldsymbol{t}$$

Equation 8.1

§ When a net force acts on an object, the impulse of the net force is equal to the change in momentum of the object.

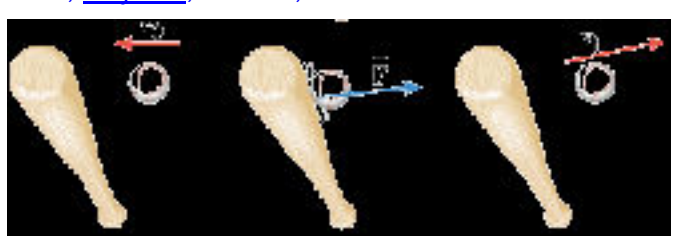
Impulse = Change in momentum

$$\boldsymbol{F}\Delta\boldsymbol{t} = \boldsymbol{mv}_f - \boldsymbol{mv}_i = \Delta\boldsymbol{mv}$$

Equation 8.2

§ It is possible for the object to undergo the same change in momentum whether it is involved in an abrupt hard collision or a cushioned collision?

» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 7, Section 7.1.



Materials

- 2 PASPORT USB Links
- PASPORT Force Sensor
- PASPORT Motion Sensor
- PAScar
- Triple-beam Scale
- 250 g Mass
- Accessory Bracket
- 1.2 m Dynamics Track
- Adjustable Feet
- Heavy Object (brick, book, etc.)

Setup *Equipment Setup*

Safety Reminder: Follow the directions for using the equipment.

§ Mount the Force Sensor on the Accessory Bracket. Mount the Accessory Bracket in the T-slot on the side of the Dynamics Track.

§ Attach the Adjustable Feet to each end of the track. Set one of the Adjustable Feet to its maximum height and set the other to its minimum height.

§ Place the Motion Sensor at the raised end of the track so it can measure the motion of the cart.

» For all trials start the cart at this place.

§ Brace the Force Sensor end of the track against a heavy object so the track will not move during the collision.

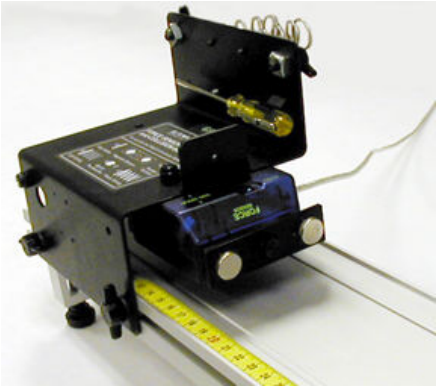
Setup

Equipment Setup

§ Place the magnetic bumper that comes with the Accessory Bracket on the front of the Force Sensor.

§ Put the 250-g mass into the tray on the the PAScar and weigh the cart and mass together. Record this mass in kilograms.

Mass of cart plus 250-g mass: _____



Force Sensor on Accessory Bracket

Equipment setup

Process *Record Data*

§ Prior to each data run, press the **Zero** button on the Force Sensor.

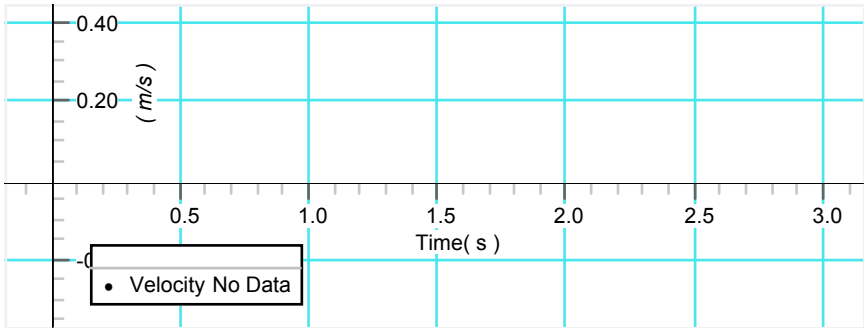
§ Hold the cart 15 cm away from the Motion Sensor.

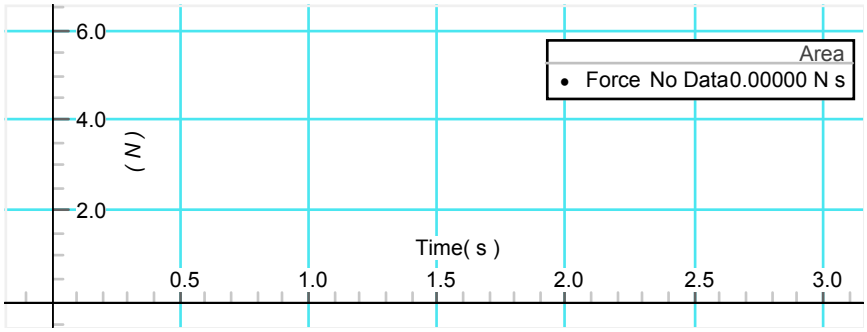
§ Click **Start**. Release the cart so that it rolls forward.

» Data recording begins when the cart collides with the bumper.

§ Click **Stop** after the PAScar has rebounded from the collision.

» The next two pages describe how to analyze your data.





Analyze *How to Find the Impulse*

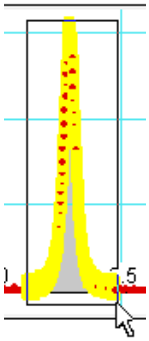
§ Select a region of the force curve.

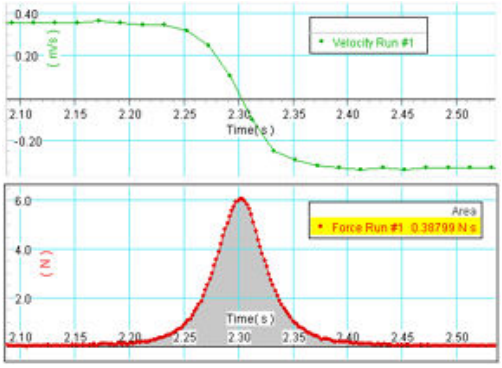
§ Right-click (or control-click) graph and choose **Scale** to resize the graph.

§ Select the region of the expanded force curve that shows the collison.

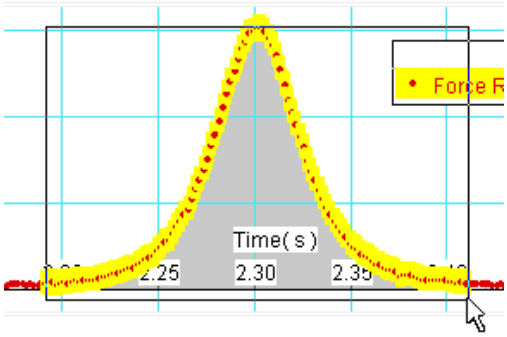
» The Legend box shows the impulse under Area. (In this example, Area is 0.37428 N s.)

Select a region to expand.

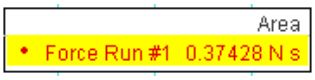




Expanded graph



Select a region that shows the collision.



Legend box

Analyze *How to Find the Change of Velocity*

§ Choose the **Smart Tool** button.

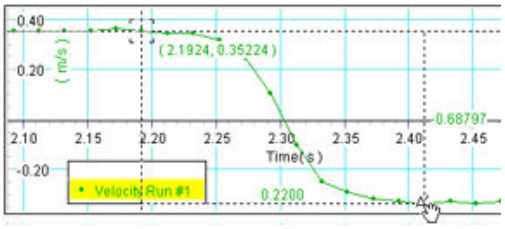
§ Drag the **Smart Tool** to the point right before the curve decreases that marks the beginning of the collision.

§ Move your cursor to the bottom right corner of the **Smart Tool**.

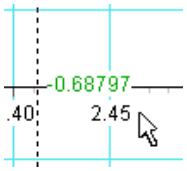
» The cursor turns into a triangle (the "delta" tool).

§ Drag the delta tool to the point at the bottom of the curve that marks the end of the collision.

» The value for the change in the Y coordinate is the change in velocity from the beginning to end of the collision. In this example, the change is -0.68797 m/s.



Drag the delta tool from the beginning to the end of the collision.



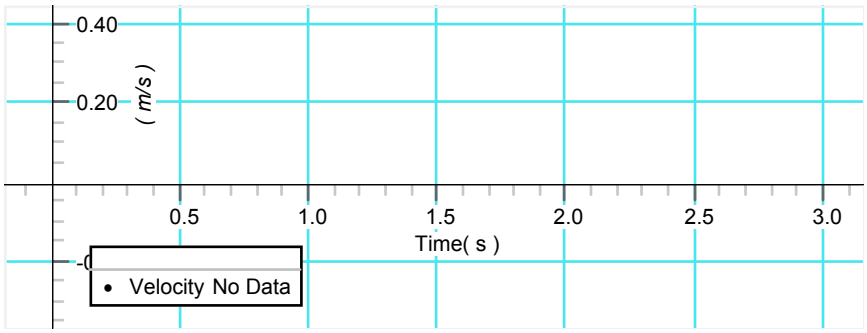
The "delta Y" value is the change in velocity for the collision.

Analyze

Find the Impulse and the Change of Velocity

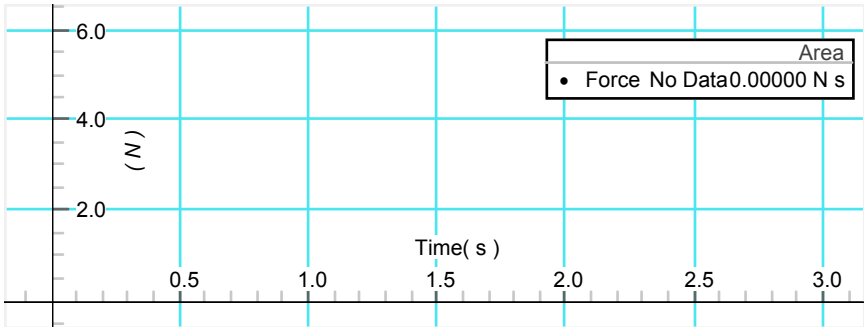
§ Use the procedure described previously to find the impulse (Area under the force curve) and the change of velocity.

§ Record the values for impulse and change of velocity.



Impulse:

Change of velocity:



Observations

§ Why does the velocity of the cart change from a positive value before the collision to a negative value after the collision?

§ Why is the area under the curve for the force graph equal to the impulse?

Analyze *Calculations and Comparisons*

§ Calculate the change in momentum. Multiply the total mass of the cart by the change in velocity. Record the result.

§ Compare the change in momentum to the impulse (Area under the force curve). Calculate the percent difference.

§ Record the percent difference.

$$\Delta mv = mv_f - mv_i$$
$$= m(v_f - v_i)$$
$$= m\Delta v$$

Cart + mass: _____

Change in velocity: _____

Change in momentum: _____

Impulse: _____

% difference: _____

$$\% \text{ difference} = \left| \frac{\Delta mv - \text{Impulse}}{\text{Impulse}} \right| x100$$

Synthesize *Conclusions and Error Analysis*

§ For your data, how does the change in momentum compare to the impulse? How does your hypothesis compare to the results?

§ What are possible reasons why the change in momentum is different from the measured impulse?

» Hypothesis:

Synthesize *Applications and Extension Problem*

§ Explain why airbags in cars can help to prevent injuries to the occupants during a frontal collision.

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 7, Problem 2, page 200.

A 62.0-kg person, standing on a diving board, dives straight down into the water. Just before striking the water, her speed is 5.50 m/s. At a time of 1.65 s after entering the water, her speed is reduced to 1.10 m/s. What is the average net force (magnitude and direction) that acts on her when she is in the water?

Notes